
SURVIVAL

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Headquarters, Department of the Army

Chapter 6

Shelter and Clothing

A shelter can protect you from the sun, insects, wind, rain, snow, hot or cold temperatures, and enemy observation. It can give you a feeling of wellbeing and help you maintain your will to survive. In some areas, your need for shelter may take precedence over your need for food and possibly even your need for water. For example, prolonged exposure to cold can cause excessive fatigue and weakness. This chapter discusses clothing and both manmade and natural shelters.

PRIMARY SHELTER

6-1. Your primary shelter in a survival situation will be your uniform. This point is true regardless of whether you are in a hot, cold, tropical, desert, or arctic situation. For your uniform to protect you, it must be in as good a condition as possible and be worn properly.

6-2. Clothing needs to keep the isolated person dry, warm or cool, out of the sun, and away from bugs. The foundation of the isolated person's uniform is their footwear. Personnel should ensure they have broken in their boots before wearing them on a mission to ensure they do not get unwanted blisters. Good socks also provide protection from blisters. Socks available to personnel are typically made of polypropylene, cotton, and wool.

6-3. The key in choosing the right sock is to ensure the material meets the isolated person's needs. For example, wool socks have great insulation value; wool is animal hair of hollow fibers and it retains heat. It absorbs moisture but retains it in its fibers, and can absorb about fifty percent of its weight in water before it feels cold and wet. Wool is hard to dry out but retains its insulating properties when wet. Wool is flame resistant and is more forgiving than other fabrics when drying by fire. The wool fiber is barbed and spear-shaped and can cause an itchy feeling. However, new manufacturing processes and different breeds of wool help with this problem.

6-4. Cotton socks keep the foot cool and wick moisture away from the foot; once wet, cotton will lose about ninety percent of its insulation value and is very difficult to dry out. In dry, warm climates, cotton socks should be the isolated person's sock of choice because they can be extremely abrasion and ultraviolet-ray resistant, and they can help keep feet cool. Polypropylene socks resist absorbing moisture. The material transfers moisture across the material from the skin onto other material or the air where it can dry out.

6-5. Polypropylene makes an excellent base layer. One down side of polypropylene is that it is so effective at wicking away moisture it can cause the body to overwork itself and use more energy than other fabrics. This fabric also retains body odor and has a low melting point, so caution should be used when drying it around a fire. Personnel should carry extra socks and rotate them as needed to keep their feet dry. Foot powder can assist in this process.

6-6. Clothing fibers can be degraded by sand and soil because of the abrasive nature of their grains. Additionally, isolated personnel may develop blisters or a hot spot causing injury when sand- or dirt-covered clothing rubs against their skin. They must make every effort to attain a high standard of cleanliness including the use of platforms to avoid putting clothing and equipment directly on the ground, but not at the expense of security.

6-7. If clothing gets dirty, clean it and keep it dry. The "COLDER" (clean, overheating, loose layers, dry, examine, and repair) principle provides a foundation for care and wear of the uniform while isolated and is further described in the following :

- **Clean.** Keep clothing clean. This principle is always important for sanitation and comfort. In winter, it is also important from the standpoint of warmth. Clothes matted with dirt and grease lose much of their insulation value. Heat can escape more easily from the body through the clothing's crushed or filled up air pockets. Dirt within the fibers will cut or tear the clothing, causing them to wear out prematurely. In a survival situation, it may be impractical to wash clothing; therefore, take the necessary precautions to prevent clothing from becoming soiled.
- **Overheating.** Avoid overheating. When personnel get too hot, they sweat and their clothing absorbs moisture. This affects their warmth in the following ways: dampness decreases the insulation quality of clothing and as sweat evaporates, the body cools. Adjust your clothing so that you do not sweat. Do this by partially opening your parka or jacket, by removing an inner layer of clothing, by removing heavy outer mittens, or by throwing back your parka hood or changing to lighter headgear. The head and hands act as efficient heat dissipaters when overheated.
- **Loose layers.** Wear clothing loose and in layers. Tight clothing and footwear restrict blood circulation and invite cold injuries. It also decreases the volume of air trapped between the layers, reducing the insulation value. Several layers of lightweight clothing are better than a single, equally thick layer of clothing, because the layers have dead airspaces between them. This dead airspace provides extra insulation. In addition, layers of clothing allow personnel to take off or add layers to prevent excessive sweating or to increase warmth.
- **Dry.** Keep clothing dry. This is important since wet clothing can conduct heat away from the body up to 25 times faster than dry clothing, depending on the type of fabric.
 - Wear water repellant clothing as an outer layer if available. It will shed most of the water from rain and snow. Despite the precautions taken, there will be times when isolated persons cannot avoid getting wet. At such times, drying their clothing may become a major problem.
 - As isolated persons move about, they can place wet clothing on their rucksack exposing the wet clothing to the sun and wind; this will dry them. They can also put damp clothing near their body's core and body heat will begin to dry the clothing. If at a stationary location, construct drying racks or a clothesline to hang clothes up to dry.
 - Placing clothing near a fire will dry it quickly. Isolated persons should make sure to use the three-second rule so they do not put clothing too close to the fire and damage it. The three-second rule involves being able to hold a hand in front of the fire at a distance that, if the back of the hand is exposed to the flame, it is still comfortable to keep it there for three or more seconds. Also, monitor the clothing to ensure that it does not fall into the flame or that the fire does not flame up and destroy the items.
 - Dry leather items slowly. If no other means are available for drying boots, isolated persons can put the boots between their sleeping bag liner and shell. Their body heat will dry the leather.
- **Examine.** Periodically examine clothing for worn areas, tears, and cleanliness.
- **Repair.** Repair any damaged clothing before tears and holes become too large to patch or sew. Patches can be made from cloth, tape, and other materials. Use a sewing kit, improvising one if needed. Sewing kits can be made from such items as bones, plant fibers, parachute cord, and large thorns.

SHELTER CONSIDERATIONS

6-8. Shelter is anything that provides protection from environmental hazards. The environment influences shelter site selection and factors, which isolated personnel must consider before constructing an adequate shelter. Depending on the environmental conditions, shelter may be the isolated person's number one priority. They can take advantage of natural occurring features or make a shelter from manmade and natural materials brought along or found in the environment. Isolated persons should consider the type of shelter, location, and time of day when looking to build a shelter. Isolated persons should also consider the location of shelter in relation to temperature, precipitation, and wind. They should consider the time and effort needed to build adequate shelter and protection from the elements. Some environments require shelter to be built off the ground, others will require more insulation, and still others will require the effective use of winds for cooling.

SHELTER SITE SELECTION

6-9. When isolated persons have prioritized shelter as a necessity, they must start looking for it as soon as possible. A good shelter site will contain enough materials to improvise the type of shelter needed as well as be large enough and level enough for the isolated person to lie down and protect their equipment. When selecting a shelter site, consider whether the site—

- Provides adequate materials to build the right type of shelter.
- Provides concealment from enemy observation.
- Provides elemental sun, wind, rain, and snow protection.
- Has camouflaged escape routes
- Is free of dead trees that may fall.
- Is located in an area free from threat of avalanches, rockslides, mudslides, Flash-flood areas and high-water marks.
- Is free from insects, reptiles, and poisonous plants. Low-lying areas that may be a cold sump.
- Is suitable for signaling and recovery.
- Contains adequate food and water.

6-10. When looking for a shelter site, isolated persons should keep in mind the type of shelter needed. They should ask themselves the following questions:

- How much time and effort will I need to build the shelter?
- Will the shelter adequately protect me from the elements?
- Do I have the tools and needed materials to complete the shelter or do I need to improvise?
- How long will I be at this site?

SITE PREPARATION

6-11. Site preparation includes brushing away rocks and twigs from the sleeping area and cutting back overhanging vegetation. Look for loose rocks, dead limbs, coconuts, or other material that could fall on the shelter. Remember thick, brushy, low ground also harbors more insects. Inspect the site for venomous snakes, ticks, mites, scorpions, and stinging ants.

CONSTRUCTION METHODS

6-12. General considerations for the construction of a shelter include the following—

- **Size.** The shelter should be large enough and require minimal effort to build.
- **Natural screens and walls.** Fallen trees, rock, large trees, etc. act as a heat reflector and aids in the dispersion of smoke from a fire.
- **Ventilation system.** The ventilation system provides fresh air, allows carbon monoxide to escape, reduces internal perspiration and acts as a heat sump.
- **Pitch and tightness.** If the shelter is built with a parachute or other porous materials, remember that "pitch and tightness" apply to shelters designed to shed rain and snow. Porous materials will not shed moisture unless they are stretched tightly at an angle of sufficient pitch, which will encourage run-off instead of penetration. An angle of 40 to 60 degrees is recommended for the pitch of the shelter. The material stretched over the framework should be wrinkle-free and tight. Do not touch the material when water is running over it as this will break the surface tension at that point and allow water to drip into the shelter.
- **Insulation.** Is added to shelters in colder environments in an effort to conserve the isolated person's loss of body heat due to conduction. Pine boughs, moss, and grasses can be used to insulate the shelter.
- **Framework.** Must be strong enough to support the weight of the covering and precipitation caused by snow or rain. It must also be sturdy enough to resist strong wind gusts.

IMMEDIATE ACTION SHELTER

6-13. An immediate action shelter is one which can be erected quickly with minimal effort. For example, from a raft, aircraft parts, parachutes, tarpaulin, or plastic material. Natural formations can also shield the isolated person from the elements immediately, to include overhanging ledges, fallen logs, caves, and tree wells. Regardless of type, the shelter must provide whatever protection is needed and, with a little ingenuity. In some instances, immediate action shelters may have to serve as permanent shelters. Examples of immediate-action shelters include—

- A wall or existing manmade structure.
- A vehicle or aircraft.
- A tarp, poncho, piece of plastic, space blanket, or evasion chart (EVC).
- Any material that is found in the area that will provide top cover.
- A cave or rock outcropping.

NATURALLY OCCURRING SHELTERS

6-14. Naturally-occurring shelters should be considered whenever it is determined that shelter is necessary, especially when isolated persons are concerned with getting out of the elements and have little time, materials, or tools. Using what is already available prevents the need for the complete construction of a shelter from scratch, and will save, time, energy, and materials (see figure 6-1). Examples of naturally-occurring shelters include—

- Caves.
- Rocky crevices.
- Small depressions.
- Large rocks on leeward sides of hills.
- Clumps of bushes.
- Large trees and low-hanging limbs.
- Fallen trees with thick branches.
- Uprooted tree buttresses.
- Tree pit snow shelters.

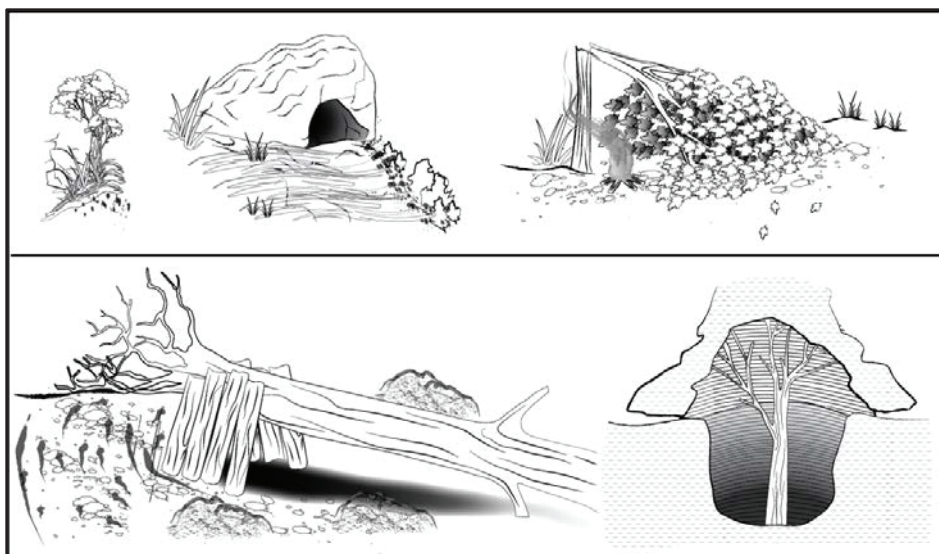


Figure 6-1. Types of naturally occurring shelters

MANMADE SHELTER CONSTRUCTION

6-15. Shelters constructed from manmade material usually require more time, energy, and materials to construct than naturally occurring shelters. However, it is best to select a site that can be improved upon with minimal effort. Existing trees can be used in various ways to cut down on the time and effort of constructing the framework. Isolated persons must gather the necessary materials to build the framework for some shelters. The manmade materials available dictate the type of shelter that can be constructed. Examples of manmade shelters include—

- Poncho lean-tos.
- Poncho tents.
- A-frames.
- Teepees.
- Para hammocks

Poncho Lean-tos

6-16. It takes only a short time and minimal equipment to build this lean-to. For the lean-to, required items are a poncho, 7 to 10 feet of rope or parachute suspension line, three stakes about 1 foot long, and two trees or two poles 7 to 10 feet apart. Before selecting the trees to be used or the location of the poles, check the wind direction. Ensure that the back of the lean-to will be into the wind (see figure 6-2, page 6-6). Use the following steps to construct a poncho lean-to:

- **Step 1.** Tie off the hood of the poncho. Pull the drawstring tight, roll the hood long ways, fold it into thirds, and tie it off with the drawstring.
- **Step 2.** Cut the rope in half. On one long side of the poncho, tie half of the rope to the corner grommet. Tie the other half to the other corner grommet.
- **Step 3.** Attach a drip stick (about a 4-inch stick) to each rope about 1 inch from the grommet. These drip sticks will keep rainwater from running down the ropes into the lean-to. Tying strings (about 4 inches long) to each grommet along the poncho's top edge will allow the water to run to and down the line without dripping into the shelter.
- **Step 4.** Tie the ropes about waist high on the trees. Use a round turn and two half hitches with a quick-release knot.
- **Step 5.** Spread the poncho and anchor it to the ground, putting sharpened sticks through the grommets and into the ground.
- **Step 6.** If the lean-to will be used for more than one night or rain is expected, make a center support for the lean-to. Make this support with a line. Attach one end of the line to the poncho hood and the other end to an overhanging branch. Make sure there is no slack in the line. Alternatively, place a stick upright under the center of the lean-to. However, this method will restrict space and movements in the shelter.
- **Step 7.** For additional protection from wind and rain, place some brush, a rucksack, or other equipment at the sides of the lean-to.
- **Step 8.** To reduce heat loss to the ground, place some type of insulating material, such as leaves or pine needles, inside the lean-to.

Note. When at rest, isolated persons lose as much as 80 percent of their body heat to the ground.

- **Step 9.** To increase security from enemy observation, lower the lean-to's silhouette by making two changes. First, secure the support lines to the trees at knee height (not at waist height) using two knee-high sticks in the two center grommets (sides of lean-to). Second, angle the poncho to the ground, securing it with sharpened sticks, as above.

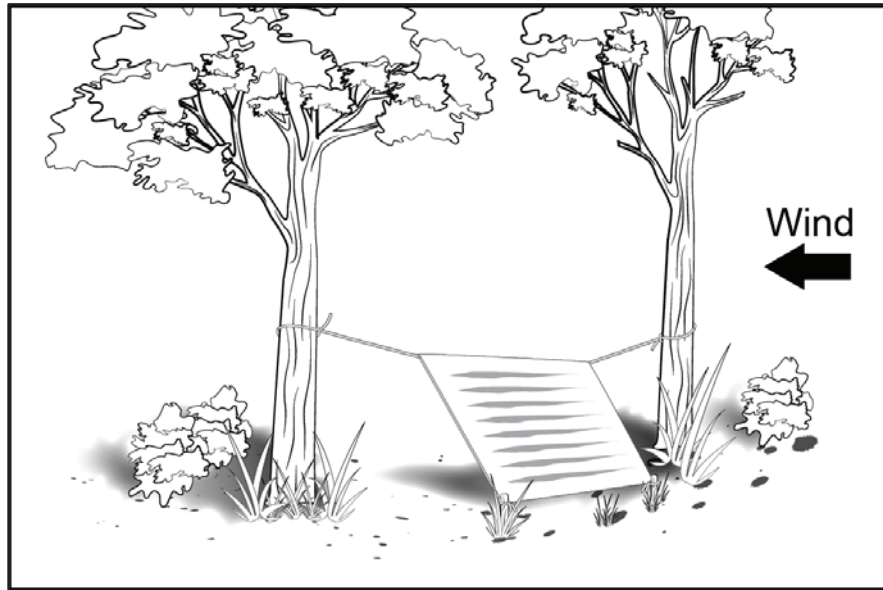


Figure 6-2. Poncho lean-to

Poncho Tent

6-17. This tent provides a low silhouette. It also protects isolated persons from the elements on two sides (see figures 6-3 and 6-4 on page 6-7). However, it has less usable space and observation area than a lean-to, decreasing reaction time to enemy detection. To make this tent, isolated persons need a poncho, two 5- to 8-foot long ropes, six sharpened sticks about 1 foot long, and two trees 7 to 10 feet apart. To construct a poncho tent, use the following steps:

- **Step 1.** Tie off the poncho hood in the same way as the poncho lean-to.
- **Step 2.** Tie a 5- to 8-foot rope to the center grommet on each side of the poncho.
- **Step 3.** Tie the other ends of these ropes at about knee height to two trees 7 to 10 feet apart and stretch the poncho tight.
- **Step 4.** Draw one side of the poncho tight and secure it to the ground pushing sharpened sticks through the grommets.
- **Step 5.** Follow the same procedure on the other side.
- **Step 6.** If a center support is needed, use the same methods as for the poncho lean-to. Another center support is an A-frame set outside but over the center of the tent. Use two 12- to 16-foot-long sticks, one with a forked end, to form the A-frame. Tie the hood's drawstring to the A-frame to support the center of the tent.

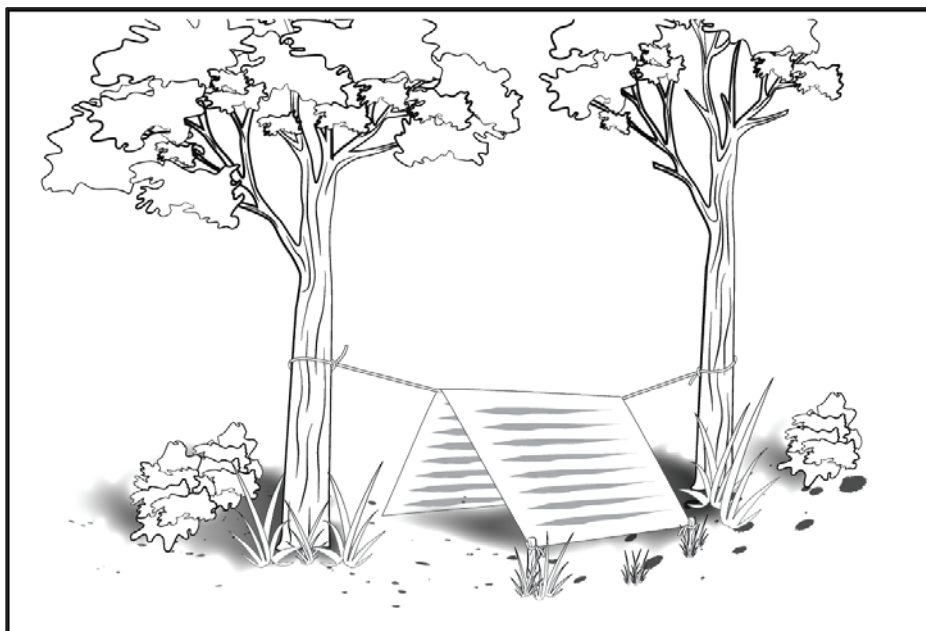


Figure 6-3. Poncho tent using overhanging branch

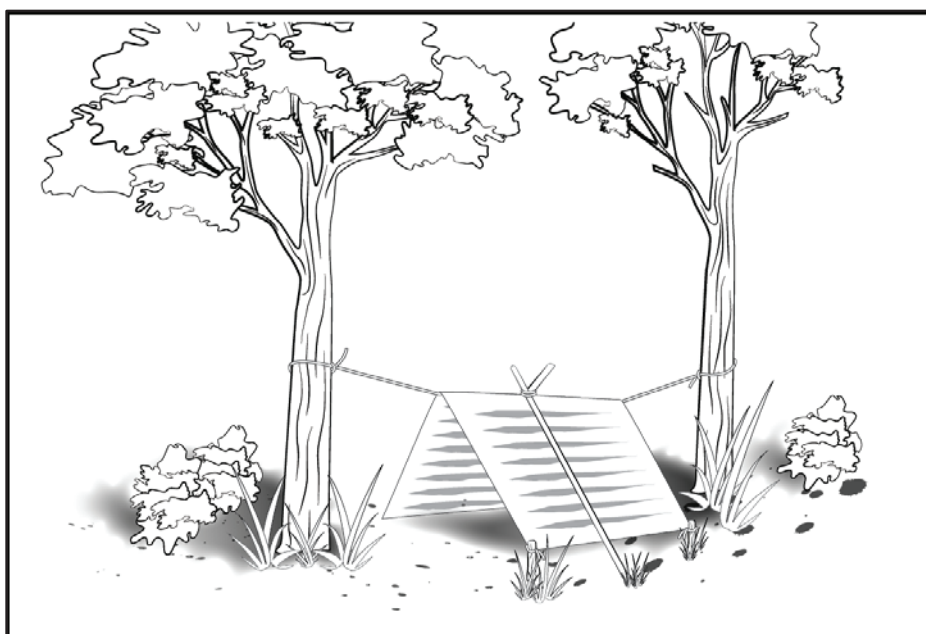


Figure 6-4. Poncho tent using center support

6-18. In open sandy terrain, soil can be placed into excess material, if available, to create pillars by stacking sand bags to elevate the outside and inside layers. The sides are left open to allow air to flow between layers (see figure 6-5 on page 6-8).

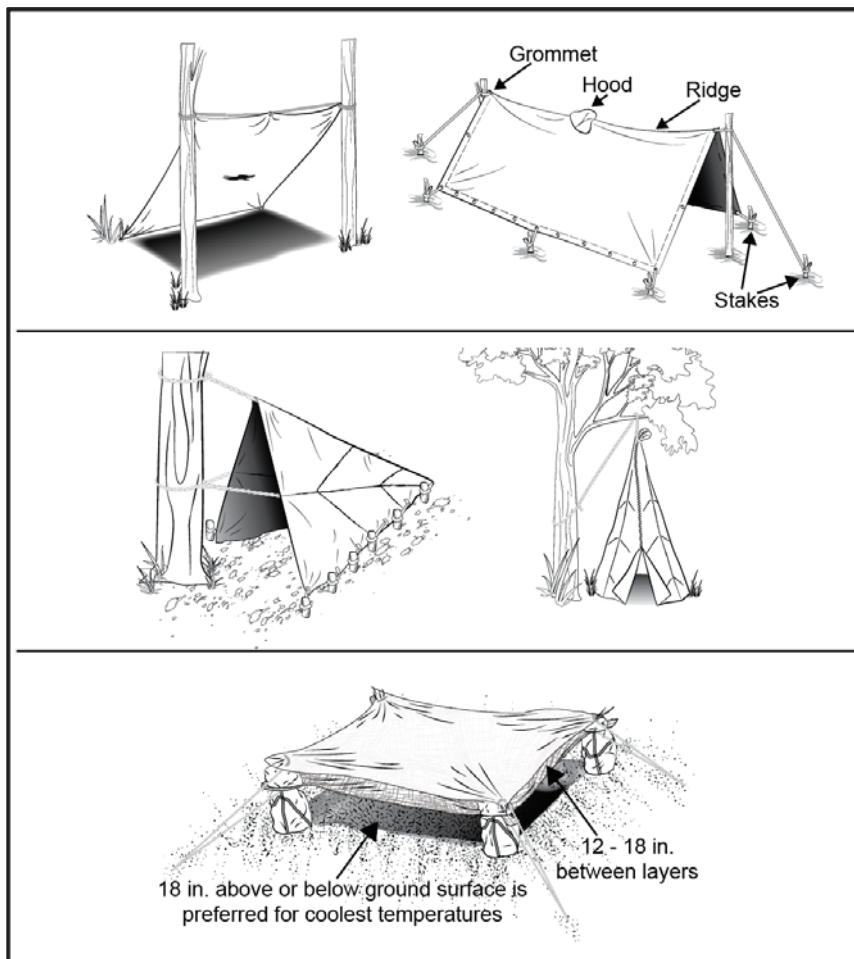


Figure 6-5. Types of manmade shelters

A-Frame Shelters

6-19. Shelter construction begins with the framework, which must be strong enough to support the weight of the covering and precipitation (ice, snow, rain). It must also be sturdy enough to resist strong wind gusts and will typically have a 40-60 degree pitch for shedding precipitation and providing shelter room. (See figure 6-6 on page 6-9.)

6-20. A-frame shelters are typically built using the following materials:

- One 12 to 18 foot long sturdy ridgepole with all projections cleaned off.
- Two bipod poles, approximately seven feet long.
- Parachute material, normally five or six gores, or other natural or manmade material
- Suspension lines or cord.
- Buttons or small objects placed behind gathers of material to provide a secure way of affixing suspension line to the material.
- Approximately 14 stakes, approximately 10 inches long.

6-21. Once the materials are collected and readied for use, the framework is assembled followed by application of the fabric. The care and techniques used to apply the covering will determine the effectiveness of the shelter in shedding precipitation. If using parachute material, stretch the center seam tight, and then work from the back of the shelter to the front, alternating sides and securing the material to stakes or

framework by using buttons and lines. Stretch the material tight by pulling the material 90 degrees to the wrinkles. If the material is not stretched tight, any moisture will pool in the wrinkles and leak into the shelter.

6-22. If natural materials are used for the covering, the shingle method should be used. Start at the bottom and work towards the top of the shelter, overlapping the bottom of each piece over the top of the preceding piece. This will allow water to drain off. Place the material on the shelter in sufficient quantity so that you cannot see through the shelter from the inside. Continue building the shelter as follows:

- Lash the two bipod poles together at eye-level height.
- Place the ridgepole, with the large end on the ground, into the bipod formed by the poles and secure with a square lash.
- The bipod structure should be 90-degrees to the ridgepole and the bipod poles should be spread out to an approximate equilateral triangle of a 60-degree pitch. A piece of line can be used to measure this.
- Tie off approximately two feet of the apex in a knot and tuck this under the butt end of the ridgepole. Use half hitches and clove hitches to secure the material to the base of the pole.
- Place the center radial seam of the parachute piece or the center of the fabric on the ridgepole. After pulling the material taut, use half hitches and clove hitches to secure the fabric to the front of the ridgepole.
- Scribe or draw a line on the ground from the butt of the ridgepole to each of bipod poles. Stake the fabric down, starting at the rear of the shelter and alternately staking from side to side to the shelter front. Use a sufficient number of stakes to ensure the parachute material is wrinkle-free.
- Stakes should be slanted or inclined away from the direction of pull. When tying off with a clove hitch, the line should pass in front of the stake first and then pass under itself to allow the button and line to be pulled 90-degrees to the wrinkle.

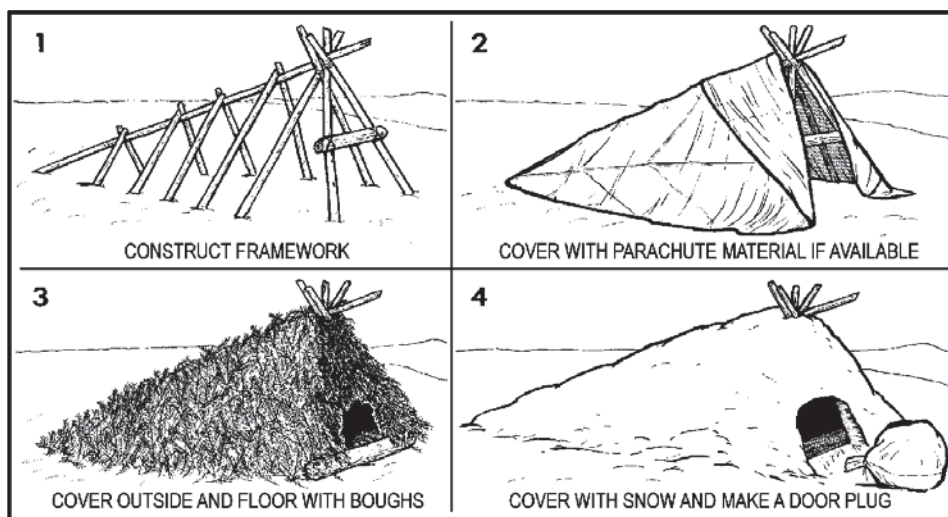


Figure 6-6. A-frame shelters

NATURAL SHELTER CONSTRUCTION

6-23. Shelters can be constructed of natural materials such as existing trees, rocks, or other useful aspects found in the environment or terrain. These natural material shelters usually require the most time and energy to construct. It is extremely important to select a site that can minimize expenditure of effort to ensure that the isolated person has the necessary energy. They may use a natural-made shelter when manmade shelter materials are limited or unavailable.

6-24. The environment or terrain will dictate what types of materials are available and ultimately what type of shelter-isolated persons will construct. All natural materials should be collected before beginning construction of the shelter. This will include the framework, covering, bedding, insulation, and materials

including lashing and staking for the shelter. If manmade lashing is unavailable, isolated persons may have to improvise cordage to be suitable. Branches may be securely placed against trees or other natural objects while support poles or branches can then be placed and/or attached depending on their function. They can use sticks with a Y fork for support or as a crossbeam. Some examples of natural made shelters are included in paragraphs 6-25—6-38.

Snow caves

6-25. The snow cave shelter (figure 6-7) is a most effective dwelling because of the insulating qualities of snow. Remember that it takes time and energy to build and that you will get wet while building it. First, you need to find a drift about three meters (10 feet) deep into which you can dig. While building this shelter, keep the roof arched for strength and to allow melted snow to drain down the sides.

6-26. Build the sleeping platform higher than the entrance. Separate the sleeping platform from the snow cave's walls or dig a small trench between the platform and the wall. This platform will prevent the melting snow from wetting you and your equipment. This construction is especially important if you have a good source of heat in the snow cave. Ensure the roof is high enough so that you can sit up on the sleeping platform.

6-27. Block the entrance with a snow block or other material and use the lower entrance area for cooking. The walls and ceiling should be at least 30 centimeters (1 foot) thick. Continue building the shelter as follows:

- Install a ventilation shaft. If you do not have a drift large enough to build a snow cave, you can make a variation of it by piling snow into a mound large enough to dig out.

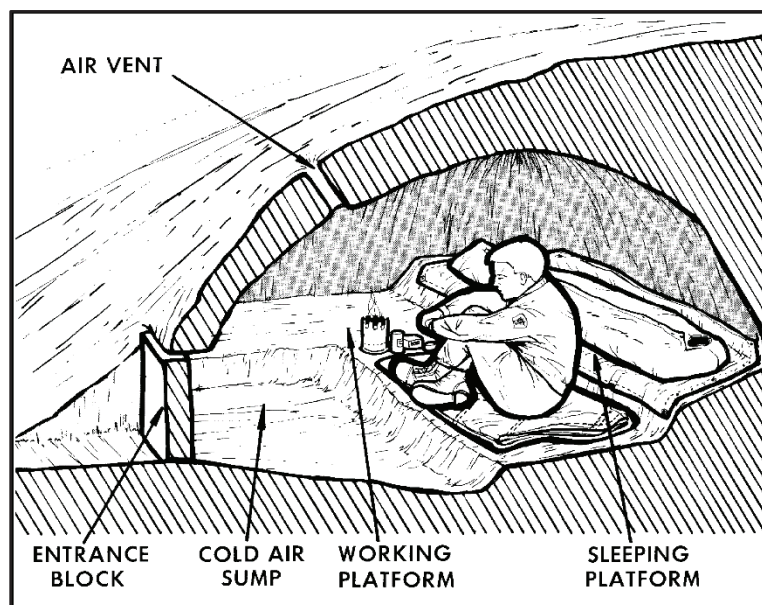


Figure 6-7 Snow Cave

Tree pit shelter

6-28. If you are in a cold, snow-covered area where evergreen trees grow and you have a digging tool, you can make a tree-pit shelter. (See figure 6-8 on page 6-11). To make this shelter, you should—

- Find a tree with bushy branches that provides overhead cover.
- Dig out the snow around the tree trunk until you reach the depth and diameter you desire, or until you reach the ground.
- Pack the snow around the top and the inside of the hole to provide support.
- Find and cut other evergreen boughs. Place them over the top of the pit to give you additional overhead cover. Place evergreen boughs in the bottom of the pit for insulation.

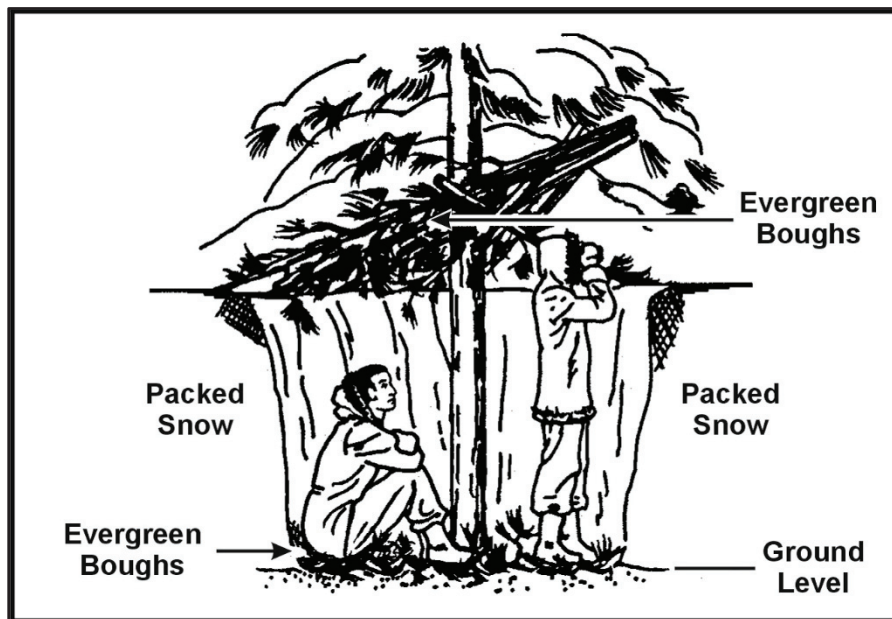


Figure 6-8 Tree-Pit Snow Shelter

Raised platforms shelter

6-29. This shelter type has many variations. One example is to use four trees or vertical poles in a rectangular pattern a little taller and a little wider than the isolated person, keeping in mind the person will also need protection for equipment. Square-lash two long, sturdy poles between the trees or vertical poles, one on each side of the intended shelter. Secure cross pieces across the two horizontal poles at six- to 12-inch intervals. This forms the platform on which a natural mattress may be constructed. Cover with an insect net of parachute material or other fabric. Figure 6-9 shows an example of a raised platform shelter.

6-30. Construct a roof over the structure using A-frame building techniques. The roof should be waterproofed with thatching laid bottom to top in a thick shingle fashion. A variation of the platform-type shelter is the para platform. A quick and comfortable bed is made by simply wrapping material around the two "frame" poles. Another method is to roll poles in the material in the same manner as for an improvised stretcher.

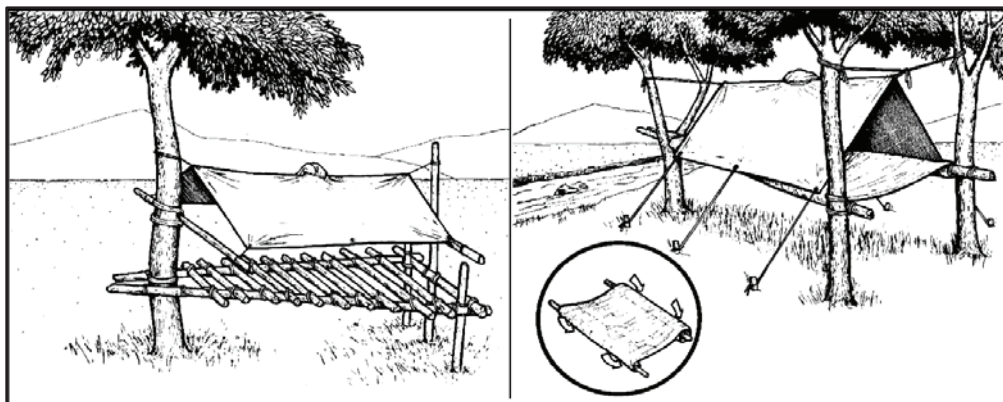


Figure 6-9. Raised Platform Shelter

Field-expedient lean-to

6-31. If you are in a wooded area and have enough natural materials, you can make a field-expedient lean-to (figure 6-10) without the aid of tools or with only a knife. It takes longer to make this type of shelter than it does to make other types, but it will protect you from the elements. To build this shelter, you will need—

- Two trees (or upright poles) about 2 meters (7 feet) apart.
- One pole about 2 meters (7 feet) long and 2.5 centimeters (1 inch) in diameter.
- Five to eight poles about 3 meters (10 feet) long and 2.5 centimeters (1 inch) in diameter for beams.
- Cord or vines for securing the horizontal support to the trees.
- Other poles, saplings, or vines to crisscross the beams.

6-32. To make this lean-to, you should—

- Tie the 2-meter (7-foot) pole to the two trees at waist to chest height. This is the horizontal support. If a standing tree is not available, construct a bipod using Y-shaped sticks or two tripods.
- Place one end of the beams (3-meter [10-foot] poles) on one side of the horizontal support. As with all lean-to type shelters, be sure to place the lean-to's backside into the wind.
- Crisscross saplings or vines on the beams.
- Cover the framework with brush, leaves, pine needles, or grass, starting at the bottom and working your way up like shingling.
- Place straw, leaves, pine needles, or grass inside the shelter for bedding.

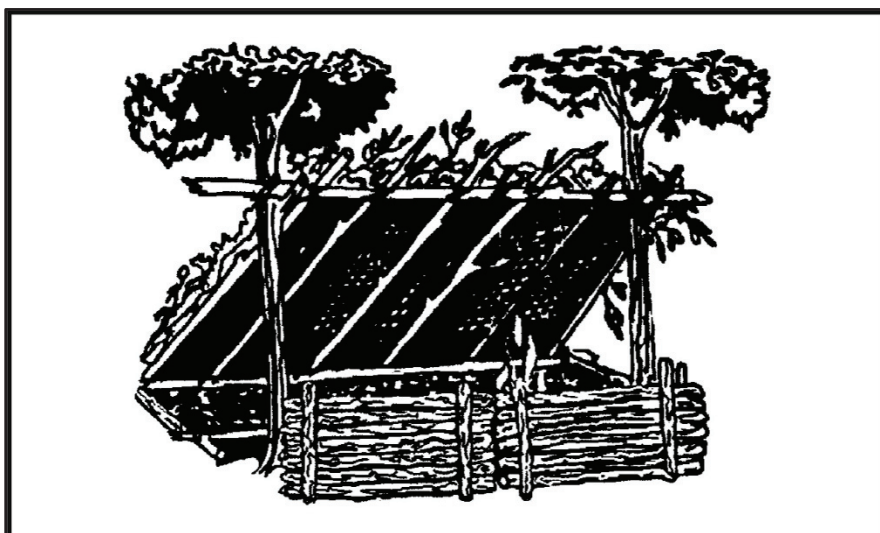


Figure 6-10- Field-expedient Lean-to and Fire Reflector

6-33. In cold weather, add to your lean-to's comfort by building a fire reflector wall. Drive four 1.5-meter-long (5-footlong) stakes into the ground to support the wall. Stack green logs on top of one another between the support stakes. Form two rows of stacked logs to create an inner space within the wall that you can fill with dirt. This action not only strengthens the wall but also makes it more heat reflective. Bind the top of the support stakes so that the green logs and dirt will stay in place.

6-34. With just a little more effort, you can have a drying rack. Cut a few 2-centimeter-diameter (3/4-inch-diameter) poles long enough to span the distance between the lean-to's horizontal support and the top of the fire reflector wall. Lay one end of the poles on the lean-to support and the other end on top of the reflector wall. Place and tie smaller sticks across these poles. You now have a place to dry clothes, meat, or fish.

Desert shelters

6-35. More complex desert shelters can be built by using the same manmade materials and incorporating new features. However, any shelter that requires significant work should be done in the cooler parts of the

day to reduce water loss and overheating. Isolated persons can add multilayered roofs that create a middle airspace to capture and dissipate the sun's heat before it reaches the living area. A middle airspace, 12 to 18 inches, can reduce the temperature in the shelter by 20 to 40 degrees Fahrenheit.

6-36. Isolated persons also can dig out the bottom of the shelter or raising the bed from the ground surface with similar effect. Digging 18 inches below the desert surface can lower the temperature in the shelter by up to 40 degrees Fahrenheit. Light colored materials if available should be used on the outside to reflect heat while the inside layer should be of dark color to filter out harmful ultraviolet rays. A below ground desert shelter is an example that incorporates these features (see figure 6-11).

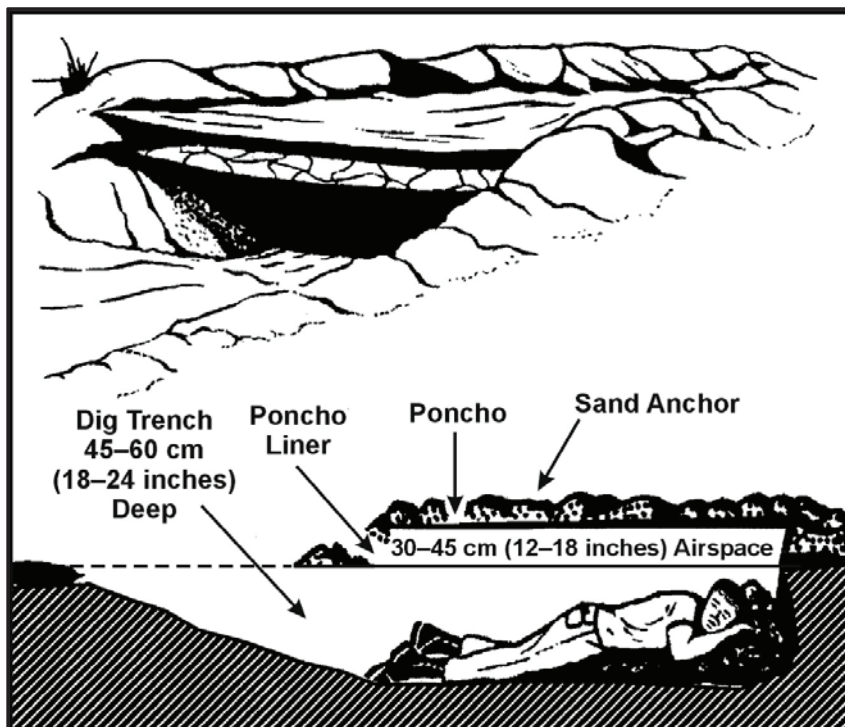


Figure 6-11- Below Ground Desert Shelter

Swamp Bed

6-37. The swamp bed is built in marshes or swamps or any area with standing water or wet ground (see figure 6-12 on page 6-14).

- Look for four trees clustered in a rectangle or cut four poles and drive them in the ground (bamboo is ideal). The four corners should form a rectangle and should be far enough apart and strong enough to support the isolated person's height and weight, including their equipment.
- Cut two poles that span the width of the rectangle. Secure the two poles to the post (or trees), making sure they are high enough to keep the bed above water and the ground to allow for high tide or high water.
- Cut additional poles, which span the rectangle's length, laying them across the two side poles and securing them. Cover the top of the bed frame with broad leaves or grass to form a softer sleeping surface. Build a fire pad by laying clay, silt, or mud at one corner of the swamp bed and allowing it to dry.

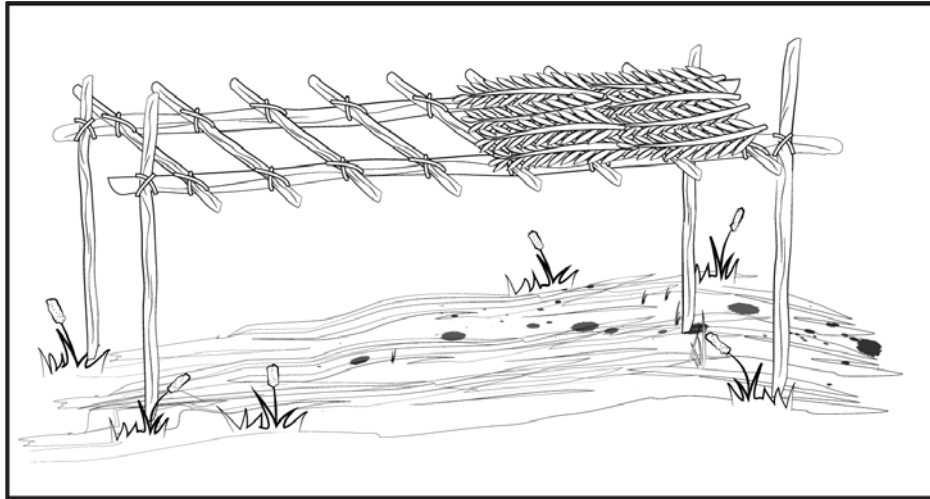


Figure 6-12- Swamp bed

Debris Hut

6-38. For warmth and ease of construction, the debris hut is one of the best (see figure 6-13 on page 6-15). When shelter is essential to survival, build this shelter. To make a debris hut, use the following steps:

- **Step 1.** Make a tripod with two short stakes and a long ridgepole or by placing one end of a long ridgepole on top of a sturdy base.
- **Step 2.** Secure the ridgepole (pole running the length of the shelter) using the tripod method or by anchoring it to a tree at about waist height.
- **Step 3.** Prop large sticks along both sides of the ridgepole to create a wedge-shaped ribbing effect. Ensure that the ribbing is wide enough to accommodate the body and steep enough to shed moisture.
- **Step 4.** Place finer sticks and brush crosswise on the ribbing. These form a latticework that will keep the insulating material (grass, pine needles, and leaves) from falling through the ribbing into the sleeping area.
- **Step 5.** Add light, dry, soft debris over the ribbing until the insulating material is at least 3 feet thick—the thicker the better.
- **Step 6.** Place a 2-foot layer of insulating material inside the shelter.
- **Step 7.** At the entrance, pile insulating material that can be dragged to the entrance from inside the shelter to close the entrance or build a door.
- **Step 8.** Finally, add shingling material or branches on top of the debris layer to prevent the insulating material from blowing away in a storm.

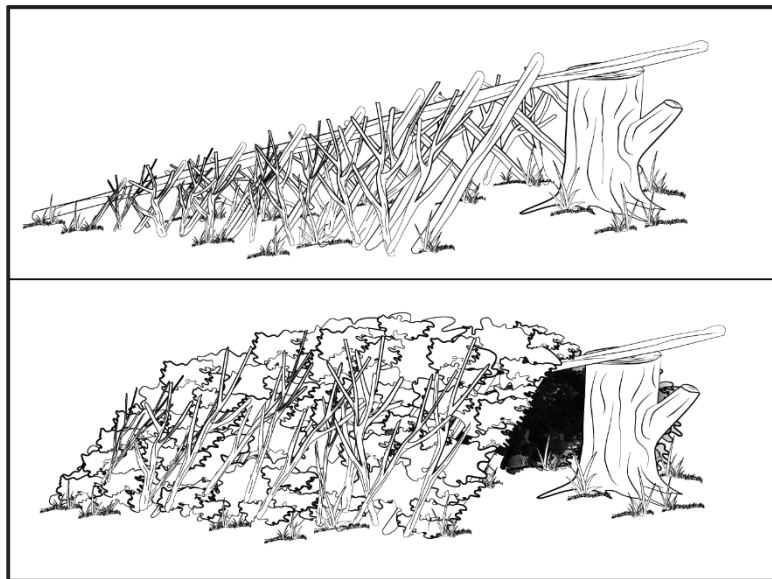


Figure 6-13. Debris hut

THERMAL PRINCIPLES AND INSULATION

6-39. A minimum of 6 to 12 inches of insulation above the isolated person is needed to retain heat. All openings except ventilation holes should be sealed to avoid heat loss. Leaving vent holes open is especially important if heat-producing devices are used (see figure 6-14, page 6-16). Candles, fuel gel, or small oil lamps produce carbon monoxide. In addition to the ventilation hole through the roof, another vent may be required at the door to ensure adequate circulation of air. As a rule, if isolated persons cannot see their breath, the snow shelter is too warm and should be cooled down to preclude melting and dripping.

6-40. If a fire is placed outside the shelter, the opening of the shelter should be placed 90 degrees to the prevailing wind. This reduces the chances of sparks and smoke being blown into the shelter if the wind reverses direction in the morning and evening, which frequently occurs in mountainous areas. The best fire-to-shelter distance is about 3 feet.

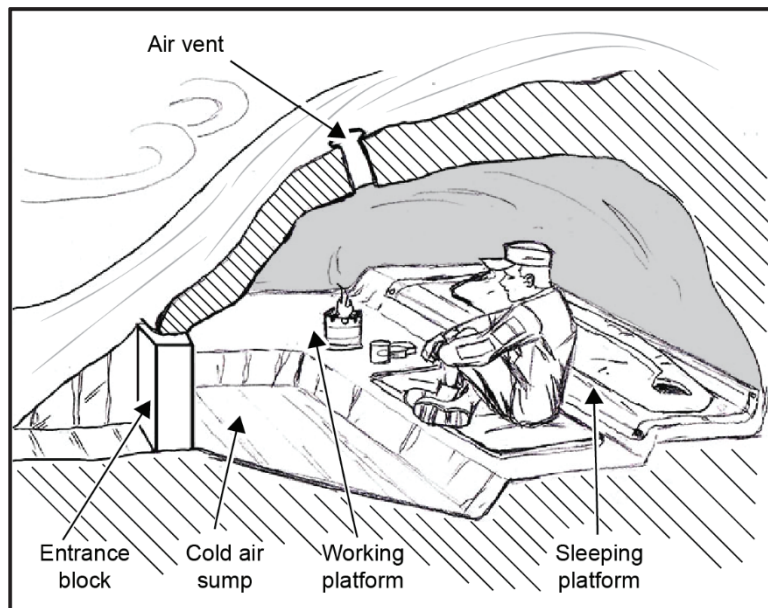


Figure 6-14. Placement of ventilation holes in a snow cave

6-41. It is unwise to build a fire in or near an aircraft or vehicle wreckage, especially if it is used as a shelter. The possibility of igniting spilled lubricants or fuels is great. Isolated persons may decide instead to use materials from the aircraft or vehicle to add to a shelter located a safe distance from the crash or abandon site.

TIE OFFS POINTS

6-42. When working with a piece of shelter covering, such as a poncho or a tarp, isolated persons can secure the material in place using the grommets that are attached to the material. However, if using a piece of plastic or parachute material, no grommets will be available. In those instances, grab a small amount of material such as dirt, pine duff, an acorn, or other similar available items about 1 inch in diameter. Grasp the piece of shelter material and place the small amount of button material on the backside of the shelter material. Gather the material around the button material making a protrusion or round bulge. Put a small staking line with a slip loop over the round bulge in the shelter material and secure the slip loop so that the line is now temporarily attached to the material. The shelter material can now be staked to the ground and any wrinkles can be pulled out of the material to create the proper pitch and tightness.

STAKING

6-43. Staking the shelter to the ground is the preferred method to achieve a safe and secure foundation. To procure stakes, sharpen several sticks 1 inch in diameter by 12 inches in length to a point, so they can be easily driven into the ground at about a 45-degree angle to provide opposing force to the shelter cover. Secure the shelter cover to the stake with a piece of line using some form of hitch. The clove hitch works well for this task.

6-44. If the ground is rocky or frozen and the sharpened stake cannot be driven into the ground, place a large-diameter branch or rock on top on the ground near the shelter covering and secure the material to the branch. The weight on the branch will achieve the same goal of securing the shelter covering in place. If ground conditions are too soft, such as in loose sand or dirt, tie the shelter covering to a small piece of wood or a rock and bury it in a hole. Cover it up with dirt to hold it in place. This type of staking is known as the “dead man staking” method.

BEDDING AND INSULATION

6-45. Heat loss through conduction can be somewhat controlled by placing a layer of insulation between isolated persons and the cold ground. This can be achieved with a manmade sleeping pad made specifically for this purpose. Ground insulation can also be improvised by taking material from the environment like vehicle seat cushions, aircraft parts, and extra clothing. If these materials are limited, use natural materials from the environment such as ferns and grass. In a coniferous forest, boughs from a tree will work.

6-46. When using natural material, make the insulation a minimum of 12 inches thick. This will allow sufficient insulation between isolated persons and the ground once the bed is compressed. The insulation should be fluffed in between uses and material added to maintain comfort.